

System Overview, This project implements two AI-based systems:

- 1. Agentic AI Customer Support System**
A LangGraph-based workflow that processes customer requests through classification, risk detection, retrieval, tool usage, response generation, memory management, and logging, with **workflow architecture design** detailed explained in readme.
- 2. Web Scraping & Summarization Pipeline**
A content processing pipeline that extracts webpage information, removes irrelevant HTML elements, handles large documents, and generates concise summaries using Gemini API with **workflow architecture design** detailed explained in readme.

Design Decisions & Trade-offs

LangGraph Workflow

LangGraph was selected to provide explicit state management and conditional routing.
Trade-off: The graph-based approach introduces additional complexity compared with a simple chain, but improves control for human escalation, tool usage, and future agent expansion.

Separated Classification and Risk Detection

Issue classification and escalation decision are handled independently.
Trade-off: Additional processing steps are introduced, but business rules remain clearer and easier to maintain.

Chunk-based Summarization

Large webpage content is divided into smaller chunks before LLM processing.
Trade-off: Long documents require more LLM calls, but this prevents context overflow and improves summarization reliability.

LLM Abstraction in Testing

Production uses Gemini API, while automated tests use an LLM interface abstraction.
Trade-off: This avoids dependency on API availability and quota limitations while maintaining deterministic validation.

Failure Handling

Failure Point	Mitigation
Security breach, data loss, outage	Human escalation workflow
Repeated customer contact	Database-based risk detection
Large webpage content	Chunk-based processing
Noisy HTML structure	Content cleaning
Excessive summary length	Output guardrail
LLM/API failure	Fallback handling

Validation

AI Customer Support Agent (7/7 test cases passed)
Web Scraping & Summarization Pipeline (5/5 test cases passed)

The system demonstrates agent orchestration, retrieval augmentation, tool integration, conversation memory, human-in-the-loop escalation, and reliable LLM-based content processing.